

Convergence of Riemannian Broyden Family of Quasi-Newton Methods with Wolfe Line Search

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Background & Motivation

- Many practical problems reduce to optimization on **Riemannian manifolds**
 - Low-rank matrix/tensor recovery ($\rightarrow$ fixed-rank manifold)
 - Subspace tracking ($\rightarrow$ Grassmann manifold)
- **Quasi-Newton methods** are popular in Euclidean optimization
 - Superlinear convergence without computing exact Hessian
 - DFP, BFGS, and their convex combination (Broyden family)
- **Goal.** Generalize the Broyden family to Riemannian manifolds and establish its global convergence with Wolfe line search.

Main Reference

- This work is mainly based on **Chapter 4** of
W. Huang, Optimization algorithms on Riemannian manifolds with applications, PhD thesis, Florida State University, 2014.
- Other key references
 - Absil, Mahony, Sepulchre — *Optimization Algorithms on Matrix Manifolds* (2008)
 - Nocedal, Wright — *Numerical Optimization* (2006)
 - Yuan — Convergence of DFP algorithm (1995)

Quasi-Newton Methods in Euclidean Space

Problem & Secant Equation

Unconstrained Optimization

$$\min_{x \in \mathbb{R}^n} f(x), \quad f \in \mathcal{C}^2$$

From Taylor expansion of ∇f at x^{k+1} , with $s^k = x^{k+1} - x^k$, $y^k = \nabla f(x^{k+1}) - \nabla f(x^k)$:

$$y^k \approx \nabla^2 f(x^{k+1}) s^k$$

Secant Equation

$$y^k = B^{k+1} s^k \quad (\text{Hessian approx.})$$

$$s^k = H^{k+1} y^k \quad (\text{inverse Hessian approx.})$$

Curvature condition. $(s^k)^\top y^k > 0$ ensures B^{k+1} can be positive definite.

Wolfe Conditions

Definition 1 (Wolfe Conditions)

For constants $0 < c_1 < c_2 < 1$ we require the step size α^k to satisfy

$$f(x^{k+1}) \leq f(x^k) + c_1 \nabla f(x^k)^\top s^k \quad (\text{Armijo})$$

$$\nabla f(x^{k+1})^\top s^k \geq c_2 \nabla f(x^k)^\top s^k \quad (\text{Curvature})$$

Lemma 2

Wolfe conditions $\implies$ curvature condition $(s^k)^\top y^k > 0$.

Proof.

$(s^k)^\top y^k = (\nabla f(x^{k+1}) - \nabla f(x^k))^\top s^k \geq (c_2 - 1) \nabla f(x^k)^\top s^k > 0$, since $c_2 < 1$ and s^k is a descent direction. □

DFP & BFGS Updates

DFP Update (Hessian approximation)

$$B^{k+1} = (I - \rho_k y^k (s^k)^\top) B^k (I - \rho_k s^k (y^k)^\top) + \rho_k y^k (y^k)^\top$$

where $\rho_k = 1/(y^k)^\top s^k$.

BFGS Update

$$B^{k+1} = B^k + \frac{y^k (y^k)^\top}{(y^k)^\top s^k} - \frac{B^k s^k (s^k)^\top B^k}{(s^k)^\top B^k s^k}$$

Both solve variational problems of the form

$$\min_B \|B - B^k\|_W \quad \text{s.t.} \quad B = B^\top, \quad B s^k = y^k$$

Broyden Family & DFP Convergence

Broyden family: convex combination of BFGS and DFP with parameter $\phi_k \in [0, 1]$:

$$B^{k+1} = (1 - \phi_k) B_{\text{BFGS}}^k + \phi_k B_{\text{DFP}}^k$$

$\phi_k = 0$: pure BFGS; $\phi_k = 1$: pure DFP.

Theorem 3 (Yuan, 1995)

If $H^0 \succ 0$, $f \in C^2$ bounded below, level set $\mathcal{L}$ convex, and

$$m\|s\|^2 \leq s^\top \nabla^2 f(x) s \leq M\|s\|^2 \quad \forall s, x \in \mathcal{L},$$

then DFP with Wolfe line search converges globally.

Riemannian Ingredients

Definition 4 (Retraction)

A smooth map $R: TM \rightarrow M$, $(p, v) \mapsto R_p(v)$, such that for the curve $c(t) = R_p(tv)$

$$c(0) = p, \quad c'(0)[v] = v \quad (\text{i.e., } dR_p(0) = \text{id}), \quad \text{for all } v \in T_p M$$

- Generalizes the **exponential map** $\exp_p$, which may be expensive to compute.
- By inverse function theorem, map R_p is a local diffeomorphism near 0.
- Computes the Riemannian gradient

$$\text{grad } f(p) = \text{grad}(f \circ R_p)(0)$$

Vector Transport

Definition 5 (Vector Transport)

A smooth map $\mathcal{T} : TM \oplus TM \rightarrow TM$, $(p, v, w) \mapsto \mathcal{T}_{v_p}(w)$, such that for some retraction R

- ① $\mathcal{T}_{v_p} : T_p M \rightarrow T_{R_p(v)} M$ is linear.
- ② $\mathcal{T}_{0_p} = \text{id}_{T_p M}$.

Two important kinds of vector transport

- **Differentiated retraction** $\mathcal{T}_R$: $\mathcal{T}_{R_{v_p}}(w) = dR_p(v)[w]$
- **Isometric vector transport** $\mathcal{T}_S$: preserves inner products

$$g_{R_p(v)}(\mathcal{T}_{S_{v_p}}(w), \mathcal{T}_{S_{v_p}}(w)) = g_p(w, w)$$

Parallel transport is a special case (isometric + geodesic).

Locking Condition

To relate $\mathcal{T}_R$ and $\mathcal{T}_S$, we impose

Definition 6 (Locking Condition)

$$\begin{aligned}\mathcal{T}_{S_\eta} \eta &= \beta \mathcal{T}_{R_\eta} \eta \\ \beta &= \frac{\|\eta\|}{\|\mathcal{T}_{R_\eta} \eta\|}\end{aligned}$$

This ensures

- $\|\mathcal{T}_{S_\eta} \eta\| = \|\eta\|$ (isometry along the search direction)
- The curvature condition $g(s^k, y^k) > 0$ is **preserved** when pushing out $\mathcal{B}^k$ via $\mathcal{T}_S$.

We also assume the operator norm bound

$$\|\mathcal{T}_{R_\eta} - \mathcal{T}_{S_\eta}\|_p \leq \delta \|\eta\|, \quad \|\mathcal{T}_{R_\eta}^{-1} - \mathcal{T}_{S_\eta}^{-1}\|_p \leq \delta \|\eta\|$$

Riemannian Secant Equation

Taylor Expansion of a Vector Field

Theorem 7 (Taylor Expansion of a Vector Field)

Let X be a smooth vector field, γ a geodesic with $\gamma(0) = p$, $\dot{\gamma}(0) = v$. Then:

$$P_{1 \rightarrow 0}^{\gamma} X_{\gamma(1)} = X_p + \sum_{i=1}^{k-1} \frac{1}{i!} \nabla_{\dot{\gamma}}^i X_p(\underbrace{v, \dots, v}_i) + R_k$$

- We pull back $X_{\gamma(t)}$ via parallel transport to $T_p M$, then apply Taylor expansion in the vector space $T_p M \cong \mathbb{R}^n$.
- Higher-order covariant derivatives along geodesics

$$\nabla^i X_{\gamma(t)}(\dot{\gamma}, \dots, \dot{\gamma}) = \underbrace{\nabla_{\dot{\gamma}} \nabla_{\dot{\gamma}} \cdots \nabla_{\dot{\gamma}}}_i X$$

Deriving the Riemannian Secant Equation

Apply the Taylor expansion to $X = \text{grad } f$ along geodesic γ_k from x^k to x^{k+1} then

$$P_{1 \rightarrow 0}^{\gamma_k} \text{grad } f(x^{k+1}) \approx \text{grad } f(x^k) + \text{Hess } f(x^k)[s_0^k]$$

Push to $T_{x^{k+1}} M$ and drop the remainder, then we similarly get the Riemannian secant equation

Riemannian Secant Equation

$$y^k = \mathcal{B}^{k+1} s^k$$

where

$$y^k = \text{grad } f(x^{k+1}) - P_{0 \rightarrow 1}^{\gamma_k} \text{grad } f(x^k)$$

$$s^k = P_{0 \rightarrow 1}^{\gamma_k} s_0^k$$

Curvature condition. $g_{x^{k+1}}(s^k, y^k) > 0$.

General Retraction & Vector Transport

In practice, we replace (since the calculation of exp and parallel transport may be expensive)

- $\exp \rightarrow$ general retraction R
- Parallel transport $\rightarrow$ isometric vector transport $\mathcal{T}_S$

Then

$$\begin{aligned}s^k &= \mathcal{T}_{S_{\alpha^k \eta^k}}(\alpha^k \eta^k) \\ y^k &= (\beta^k)^{-1} \text{grad } f(x^{k+1}) - \mathcal{T}_{S_{\alpha^k \eta^k}}(\text{grad } f(x^k))\end{aligned}$$

where $\beta^k = \frac{\|\alpha^k \eta^k\|_{x^k}}{\|\mathcal{T}_{R_{\alpha^k \eta^k}}(\alpha^k \eta^k)\|_{x^{k+1}}}$, and $\eta^k = -(\beta^k)^{-1} \text{grad } f(x^k)$ is the search direction.

Riemannian Broyden Update

DFP & BFGS on Manifolds

DFP update. solve variational problem with operator norm $\|\cdot\|_W$ on manifold

$$\mathcal{B}^{k+1} = \left(\text{id} - \frac{y^k(s^k)^\flat}{g(y^k, s^k)} \right) \tilde{\mathcal{B}}^k \left(\text{id} - \frac{s^k(y^k)^\flat}{g(y^k, s^k)} \right) + \frac{y^k(y^k)^\flat}{g(y^k, s^k)}$$

BFGS update.

$$\mathcal{B}^{k+1} = \tilde{\mathcal{B}}^k + \frac{y^k(y^k)^\flat}{g(y^k, s^k)} - \frac{\tilde{\mathcal{B}}^k s^k (\tilde{\mathcal{B}}^k s^k)^\flat}{g(s^k, \tilde{\mathcal{B}}^k s^k)}$$

Here $\flat : T_p M \rightarrow T_p^* M$ is the musical isomorphism defined by $v^\flat = g(v, \cdot)$, and

$$\tilde{\mathcal{B}}^k = \mathcal{T}_{S_{\alpha^k \eta^k}} \circ \mathcal{B}^k \circ \mathcal{T}_{S_{\alpha^k \eta^k}}^{-1}$$

is the push-out of $\mathcal{B}^k$ from $T_{x^k} M$ to $T_{x^{k+1}} M$.

Euclidean vs. Riemannian — Notation

	Euclidean	Riemannian
Inner product	$(s^k)^\top y^k$	$g(s^k, y^k)$
Dual pairing	$u v^\top$	$u v^b$
Identity	I	id
Previous B^k	B^k (reuse directly)	$\tilde{B}^k$ (push-out via $\mathcal{T}_S$)

Key difference. In $\mathbb{R}^n$, all tangent spaces are the same, so B^k can be reused directly. On manifolds, $B^k : T_{x^k}M \rightarrow T_{x^k}M$ must first be **transported** to $T_{x^{k+1}}M$ by

$$\tilde{B}^k = \mathcal{T}_S \circ B^k \circ \mathcal{T}_S^{-1}$$

After this push-out, the update formulas are **formally identical**.

Euclidean vs. Riemannian — BFGS Update

Euclidean BFGS.

$$B^{k+1} = B^k + \frac{y^k (y^k)^\top}{(y^k)^\top s^k} - \frac{B^k s^k (s^k)^\top B^k}{(s^k)^\top B^k s^k}$$

Riemannian BFGS.

$$\mathcal{B}^{k+1} = \tilde{\mathcal{B}}^k + \frac{y^k (y^k)^\flat}{g(y^k, s^k)} - \frac{\tilde{\mathcal{B}}^k s^k (\tilde{\mathcal{B}}^k s^k)^\flat}{g(s^k, \tilde{\mathcal{B}}^k s^k)}$$

Correspondence.

- $B^k \longrightarrow \tilde{\mathcal{B}}^k$ (push-out replaces direct reuse)
- $(\cdot)^\top \longrightarrow (\cdot)^\flat$ (transpose $\rightarrow$ musical isomorphism)
- $u^\top v \longrightarrow g(u, v)$ (dot product $\rightarrow$ Riemannian metric)

Broyden Family on Manifolds

Riemannian Broyden Update

$$\begin{aligned} \mathcal{B}^{k+1}(\cdot) = & \tilde{\mathcal{B}}^k(\cdot) + \frac{y^k(y^k)^b(\cdot)}{g(y^k, s^k)} - \frac{\tilde{\mathcal{B}}^k s^k (\tilde{\mathcal{B}}^k s^k)^b(\cdot)}{g(s^k, \tilde{\mathcal{B}}^k s^k)} \\ & + \phi_k g(s^k, \tilde{\mathcal{B}}^k s^k) v^k (v^k)^b(\cdot) \end{aligned}$$

where $v^k = \frac{y^k}{g(y^k, s^k)} - \frac{\tilde{\mathcal{B}}^k s^k}{g(s^k, \tilde{\mathcal{B}}^k s^k)}$.

- $\phi_k = 0$: Riemannian BFGS
- $\phi_k = 1$: Riemannian DFP
- $\phi_k \in [0, 1]$: Riemannian Broyden family

Wolfe Line Search on Manifolds

Given search direction $\eta^k = -(\mathcal{B}^k)^{-1} \text{grad } f(x^k)$ and step size α^k

Riemannian Wolfe Conditions ($0 < c_1 < c_2 < 1$)

$$f(R_{x^k}(\alpha^k \eta^k)) \leq f(x^k) + c_1 \alpha^k g_{x^k}(\text{grad } f(x^k), \eta^k)$$

$$\left. \frac{d}{dt} f(R_{x^k}(t\eta^k)) \right|_{t=\alpha^k} \geq c_2 \left. \frac{d}{dt} f(R_{x^k}(t\eta^k)) \right|_{t=0}$$

Algorithm

Riemannian Broyden's Method

Require: (M, g) , retraction R , isometric transport $\mathcal{T}_S$, $f \in C^2$, $x^0 \in M$, $\mathcal{B}^0 \succ 0$, $\{\phi_k\} \subset [0, 1]$,
 $0 < c_1 < c_2 < 1$, $\varepsilon > 0$.

1: $k \leftarrow 0$

2: **while** $\|\text{grad } f(x^k)\|_{x^k} > \varepsilon$ **do**

3: Compute search direction: $\eta^k = -(\mathcal{B}^k)^{-1} \text{grad } f(x^k)$

4: Find step size α^k satisfying Riemannian Wolfe conditions

5: Update: $x^{k+1} = R_{x^k}(\alpha^k \eta^k)$

6: Compute:

$$s^k = \mathcal{T}_{S_{\alpha^k \eta^k}}(\alpha^k \eta^k)$$

$$y^k = (\beta^k)^{-1} \text{grad } f(x^{k+1}) - \mathcal{T}_{S_{\alpha^k \eta^k}}(\text{grad } f(x^k))$$

7: **end while**

Riemannian Broyden's Method

7: Push out: $\tilde{\mathcal{B}}^k = \mathcal{T}_{S_{\alpha^k \eta^k}} \circ \mathcal{B}^k \circ \mathcal{T}_{S_{\alpha^k \eta^k}}^{-1}$

8: Broyden update

$$\begin{aligned}\mathcal{B}^{k+1}(\cdot) = \tilde{\mathcal{B}}^k(\cdot) &+ \frac{y^k(y^k)^\flat(\cdot)}{g(y^k, s^k)} - \frac{\tilde{\mathcal{B}}^k s^k (\tilde{\mathcal{B}}^k s^k)^\flat(\cdot)}{g(s^k, \tilde{\mathcal{B}}^k s^k)} \\ &+ \phi_k g(s^k, \tilde{\mathcal{B}}^k s^k) v^k (v^k)^\flat(\cdot)\end{aligned}$$

9: $k \leftarrow k + 1$

Convergence Analysis

Key Assumptions

Assumption 1 (Smoothness)

f is twice continuously differentiable on M .

Assumption 2 (Retraction domain)

$\exists r, \rho > 0$ such that for each y in the retraction neighborhood

$$\tilde{\Omega}_\rho = \{R_p(t\eta) : p \in \Omega_{x_0}, \|\eta\| = 1, t \in [0, r]\},$$

R_y is a diffeomorphism on $B(0, \rho) \subset T_y M$ and $\tilde{\Omega} \subset R_y(B(0, \rho))$.

Assumption 3 (Strong retraction-convexity)

For $t \in [0, \alpha^k]$, $R_{x^k}(t\eta^k) \in \tilde{\Omega}_{x^k}$. The function f is **strongly retraction-convex** w.r.t. R in $\overline{\tilde{\Omega}_{x_0}}$.

Retraction-Convexity

Definition 8

f is *retraction-convex* w.r.t. R in $S \subseteq M$ if

$$t \mapsto f(R_p(t\eta)) \text{ is convex}$$

for all $p \in S$, $\|\eta\| = 1$, with $R_p(t\eta) \in S$.

Lemma 9

Under Assumptions 1–2, with $R = \exp$

$$f \text{ retraction-convex} \iff \text{Hess } f \succeq 0$$

$$f \text{ strongly retraction-convex} \iff \text{Hess } f \succeq \lambda_0 \text{id}, \lambda_0 > 0$$

This connects the intrinsic Hessian to the 1D convexity of retraction curves.

Curvature Condition is Preserved

Lemma 10

Under the locking condition, Algorithm produces well-defined iterates with $\mathcal{B}^k \succ 0$ for all k , and

$$g(s^k, y^k) \geq (c_2 - 1)\alpha^k g(\text{grad } f(x^k), \eta^k) > 0$$

Proof sketch.

- 1 Define $\tilde{m}_k(t) = f(R_{x^k}(t\eta^k/\|\eta^k\|))$.
- 2 Use locking condition to relate $\mathcal{T}_S$ and $\mathcal{T}_R$ terms.
- 3 Apply Wolfe curvature condition to get the lower bound.
- 4 Positive definiteness of $\mathcal{B}^{k+1}$ is proved by induction using the determinant formula and Cauchy–Schwarz.

Preliminary Estimates (I)

Lemma 11

$$\frac{1}{2}a_0\|s^k\|^2 \leq (c_1 - 1)\alpha^k g(\text{grad } f(x^k), \eta^k)$$

From Taylor expansion + Armijo condition.

Lemma 12 (Dennis–Moré type bound)

$\exists 0 < b_0 < b_1$ such that

$$b_0 \|s^k\|^2 \leq g(s^k, y^k) \leq b_1 \|s^k\|^2$$

From integral representation of $g(s^k, y^k)$ and strong retraction-convexity bounds.

Preliminary Estimates (II)

Lemma 13

$\exists 0 < a_2 < a_3$ such that

$$a_2 \|\operatorname{grad} f(x^k)\| \cos \theta_k \leq \|s^k\| \leq a_3 \|\operatorname{grad} f(x^k)\| \cos \theta_k$$

where $\cos \theta_k = \frac{-g(\operatorname{grad} f(x^k), \eta^k)}{\|\operatorname{grad} f(x^k)\| \|\eta^k\|}$.

This tells us, **the step size $\|s^k\|$ is controlled by the gradient norm and the search direction quality.**

Preliminary Estimates (III)

Lemma 14 (Zoutendijk-type condition)

$\exists a_5 > 0$ such that

$$f(x^{k+1}) - f(x^*) \leq (1 - a_5 \cos^2 \theta_k)(f(x^k) - f(x^*))$$

Key consequence. whenever $\cos \theta_k \geq \delta_0 > 0$, we get a **contraction**.

Product of Step Sizes

Lemma 15

Under Assumptions 1–3 with $\phi_k \in [0, 1]$, $\exists a_{13} > 0$ such that

$$\prod_{j=1}^k \alpha^j \geq a_{13}^k \quad \forall k \geq 1$$

Proof sketch. (the $\hat{B}_k$ are the coordinate representations of $\tilde{B}^k$ in an orthonormal basis)

- ① **Trace bound.** $\text{trace}(\hat{B}_k) \leq b_0^k$.
- ② **Determinant bound.** $\det(\hat{B}_{k+1}) \geq \det(\hat{B}_k) \cdot \frac{g(y^k, s^k)}{g(s^k, \tilde{B}^k s^k)}$.
- ③ **AM–GM.** Eigenvalue arithmetic-geometric mean inequality

$$\det(\hat{B}_{k+1}) \leq \left(\frac{\text{tr}(\hat{B}_{k+1})}{d} \right)^d$$

- ④ Combine to get geometric lower bound for $\prod \alpha^j$.

Main Theorem

Theorem 16 (Main Result)

Suppose Assumptions 1–3 hold and $\phi_k \in [0, 1 - \delta]$ for some fixed $\delta > 0$. Then the sequence $\{x^k\}$ generated by the Riemannian Broyden method converges to the minimizer x^ .*

Key requirement: $\phi_k \leq 1 - \delta$ (bounded away from pure DFP).

- Pure BFGS ($\phi_k = 0$): always covered.
- Pure DFP ($\phi_k = 1$): **not** covered by this theorem.

Proof Strategy

Assume $\cos \theta_k \rightarrow 0$ (search direction deteriorates). Then

- 1 The refined trace inequality

$$\text{tr}(\hat{B}_{k+1}) \leq \text{tr}(\hat{B}_k) + a_8 + t_k \alpha^k$$

$$\text{where } t_k = \phi_k a_8 a_{10} - \frac{a_{11}(1-\phi_k)}{\cos^2 \theta_k} + \frac{2\phi_k a_{12}}{\cos \theta_k}.$$

- 2 Since $\phi_k \leq 1 - \delta$, the dominant term is $-\frac{a_{11}\delta}{\cos^2 \theta_k} \rightarrow -\infty$.
- 3 So $t_k \rightarrow -\infty$, specifically $t_k < -\frac{2a_8}{a_{13}}$ for k large.

Proof Strategy

- 4 By AM–GM and the product bound $\prod \alpha^j \geq a_{13}^k$

$$\sum_{j=1}^k \alpha^j \geq k a_{13}$$

- 5 Summing the trace inequality from K_0 to k :

$$\begin{aligned} 0 < \operatorname{tr}(\hat{B}_{k+1}) &\leq \operatorname{tr}(\hat{B}_{K_0}) + a_8(k+1-K_0) - \frac{2a_8}{a_{13}} \sum_{j=K_0}^k \alpha^j \\ &\leq \operatorname{const} - a_8 k \rightarrow -\infty \end{aligned}$$

Contradiction! since $\operatorname{tr}(\hat{B}_{k+1}) > 0$.

Proof Strategy

Therefore $\cos \theta_k \not\rightarrow 0$, i.e., $\exists \delta_0 > 0$ and subsequence $\{k_j\}$ with

$$\cos \theta_{k_j} > \delta_0$$

By the Zoutendijk-type lemma

$$f(x^{k_j+1}) - f(x^*) \leq (1 - a_5 \delta_0^2) (f(x^{k_j}) - f(x^*))$$

Since $\{f(x^k)\}$ is non-increasing (Armijo) and bounded below

$$\boxed{f(x^k) \rightarrow f(x^*), \quad x^k \rightarrow x^*}$$

Thank You!